

ENDG 319



Probability, Statistics and Machine Learning

Lecture 1: Data, Sampling, and Statistical Thinking

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Why engineers need statistics

Engineering decisions are often made with incomplete, noisy, or variable data.

- Repeated scientific measurements are not identical.
- Product quality, process performance, and reliability all vary.
- A sample is usually much smaller than the system we care about.
- Statistical reasoning turns data into defensible conclusions.

Statistics is the engineering language for uncertainty.

Learning outcomes for Lecture 1

BY THE END OF TODAY, STUDENTS SHOULD BE ABLE TO

- Distinguish a population, sample, parameter, and statistic.
- Explain why different random samples produce different results.
- Recognize sampling bias and the limits of convenience sampling.
- Classify variables and describe a dataset in engineering terms.
- Compare controlled experiments with observational studies.

Core mindset: use limited data to learn about a larger system.

The central question

How can engineers learn about a large or uncertain system from limited data?



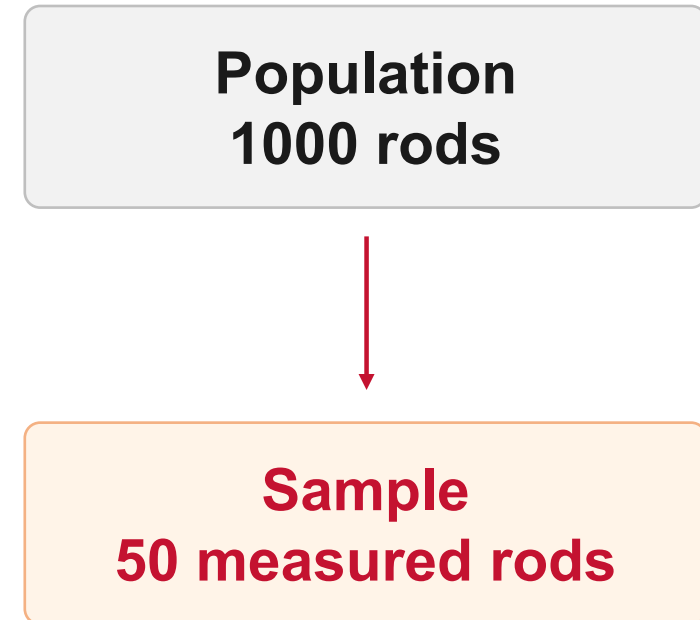
The course develops the tools needed to move responsibly along this chain.

A first engineering problem

Steel rods must satisfy a diameter specification:

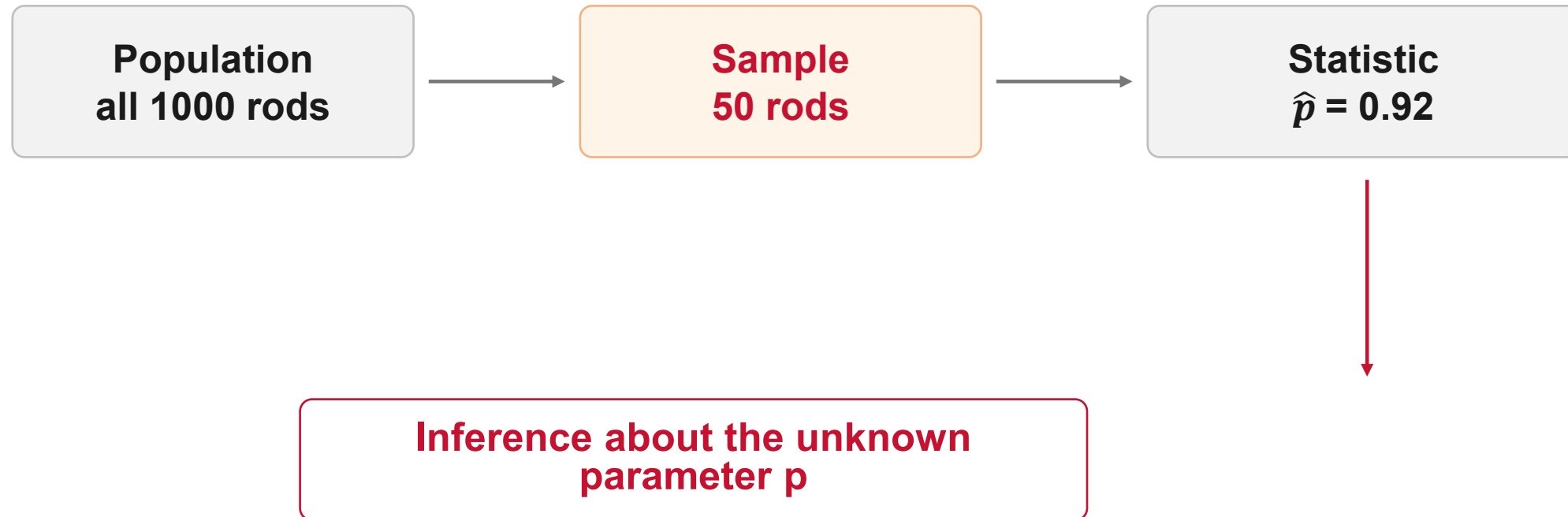
$$D = 0.45 \pm 0.02 \text{ cm}$$

- A machine produces a batch of 1000 rods.
- Inspecting every rod may be slow or expensive.
- An engineer randomly selects 50 rods.
- 46 of the 50 rods meet specification.



Observed pass rate: $46/50 = 0.92$

Population, sample, parameter, statistic



Parameter: a numerical characteristic of the population.
Statistic: a numerical summary computed from the sample.

Definitions used throughout the course

Population

Entire collection of objects or outcomes about which information is sought.

Sample

Subset of the population that is actually observed.

Parameter

Numerical characteristic of the population; usually unknown.

Statistic

Numerical summary calculated from the sample; observed from data.

A statistical problem starts by identifying these four pieces.

Simple random samples

A simple random sample (SRS) of size n is selected so that every possible collection of n population items is equally likely to be chosen.

SMALL EXAMPLE

Population = {Andy, Brian, Cole}
Select a sample of size 2

{Andy, Brian}

{Andy, Cole}

{Brian, Cole}

Each of the three possible samples has the same probability.

Collecting an SRS in practice

Example: select 10 students from a class of 100.

Student	ID

Andy	1
Jane	2
Mila	3
...	...
Brian	100



Random ID selection			

87	4	16	32
53	20	10	72
36	70		

Randomization protects against selection bias, but it does not eliminate sampling variation.

Sampling variation

Two different samples from the same population will usually be different.

SRS #1	SRS #2
96	25
68	14
1	92
90	65
29	29
7	69
25	76
53	43
3	3
57	60

Sampling variation

Natural sample-to-sample differences
caused by random selection.

**Key idea: the statistic changes from
sample to sample.**

What sampling variation means for the rods

In the sample, 46 out of 50 rods met the specification:

$$\hat{p} = 46/50 = 0.92$$

- The population proportion p is unknown.
- It is unlikely that p is exactly 0.92.
- A different random sample of 50 rods could give a different $\hat{p}$.
- Later in the course, we will quantify how close $\hat{p}$ is likely to be to p .

For today, the goal is to understand the structure of the problem.

Sampling variation is not sampling bias

Sampling variation

Random differences among properly selected samples.

Can usually be reduced by increasing sample size.

Sampling bias

Systematic error caused by how observations are selected or measured.

A larger sample does not necessarily fix it.

Random error causes scatter. Bias causes systematic distortion.

Convenience samples

A convenience sample is obtained in an easy way, not by a well-defined random method.

EXAMPLE

A construction engineer receives 1000 concrete blocks and wants to measure crushing strength in 10 blocks.

**Convenient choice:
Take 10 blocks from the top**

**Statistical concern:
Top blocks may differ from hidden
blocks**

Use convenience samples only when random sampling is not feasible, and always discuss possible bias.

Independence in sampling

Sampled observations are independent if knowing some values does not help predict the others.

Sampling with replacement

Exact independence

Sampling without replacement

Technically dependent

Practical approximation:

When the population is finite but the sample is small relative to the population, observations are often treated as approximately independent. A common rule is to be cautious when the sample exceeds about 5% of the population.

Conceptual populations

Not every population is a physical collection of objects.

- A geologist weighs the same rock several times on a sensitive scale.
- Each reading is slightly different because of measurement variation.
- The population can be viewed as all readings the scale could produce in principle.

Engineering version: all pressure readings a reactor could produce under nominally identical conditions.

Conceptual populations are essential for repeated measurements and experimental data.

Variables: what are we measuring?

A variable is a characteristic that can take different values for different objects or outcomes.

Quantitative variables

Temperature
Pressure
Tensile strength
Particle diameter
Conversion

Categorical variables

Material grade
Catalyst type
Pass/fail result
Failure mode
Manufacturing line

Important: a number is not necessarily quantitative. Student ID, postal code, or equipment ID may be categorical labels.

What is a dataset?

A dataset usually has rows as observational units and columns as variables.

Rod ID	Diameter(cm)	Pass?	Machine
R-001	0.452	Yes	A
R-002	0.468	Yes	A
R-003	0.431	Yes	A
R-004	0.476	No	A
R-005	0.448	Yes	A

Rows
individual rods

Columns
variables

Before analyzing data, identify the observational unit and the variables.

How engineering data are generated

One-sample study

One population of interest; one sample is drawn from it.

Multi-sample study

Two or more populations; one sample is drawn from each.

Example comparison:

One machine producing rods vs. two machines producing rods under different conditions.

This distinction determines which statistical method is appropriate later.

Controlled experiments vs. observational studies

Controlled experiment

The investigator controls or assigns factor values.

A well-designed randomized experiment can provide strong evidence for cause and effect.

Observational study

The investigator observes existing factor values without assigning them.

Associations may be affected by confounding variables.

Causality requires design, not just data volume.

Looking ahead: what can we learn from a sample?

Rod example: $\hat{p} = 0.92$ from a sample of 50 rods.

- How close is $\hat{p}$ likely to be to the true population proportion p ?
- How wide should a reasonable interval for p be?
- Can we conclude that at least 90% of the rods meet specification?
- How should sample size affect our certainty?

These questions lead to sampling distributions, confidence intervals, and hypothesis tests.

Key takeaways

- A sample provides information about a population, but sample results vary.
- A statistic is computed from data; a parameter describes the population.
- Random sampling reduces selection bias but does not eliminate sampling variation.
- Convenience sampling can introduce systematic bias.
- The way data are generated determines what conclusions are justified.

Next: organizing and summarizing quantitative data.